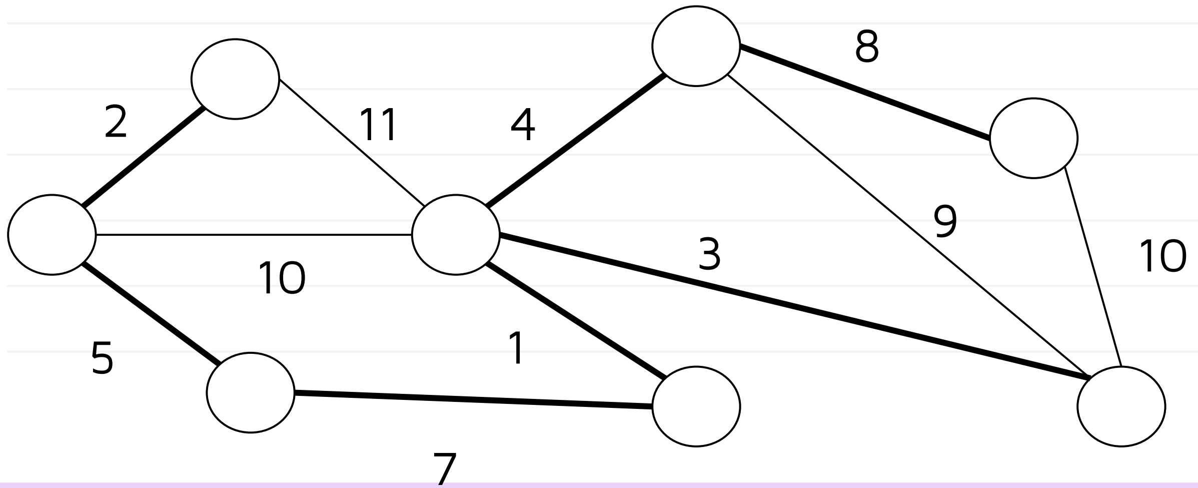
DSC 40B

Lecture 29 : Minimum Spanning Trees. Kruskal's algorithm

Minimum Spanning Trees

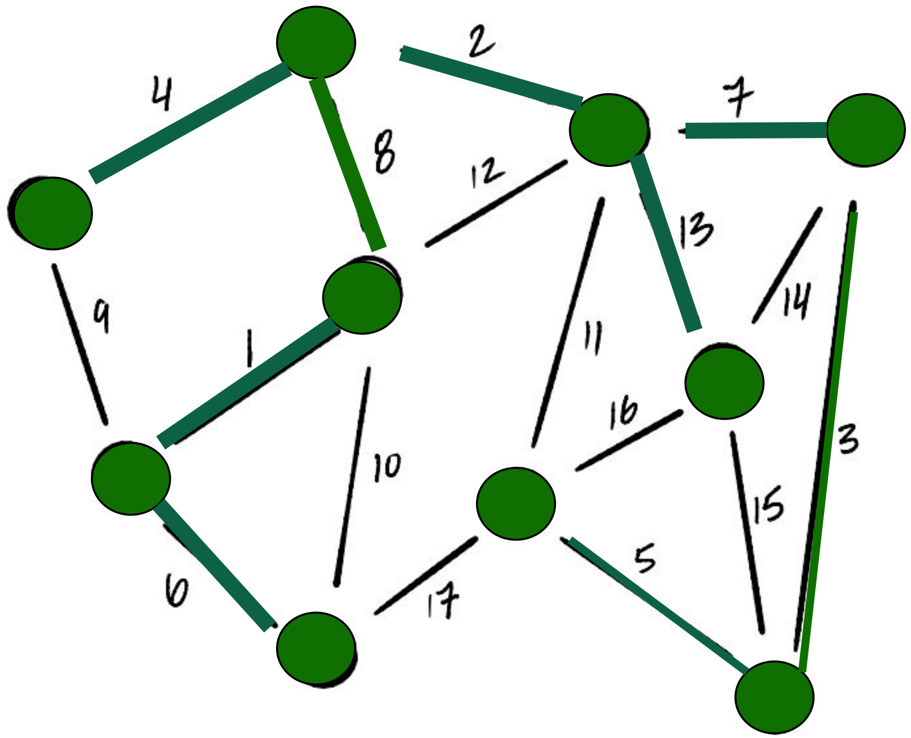
Last time: Minimum Spanning Tree

- The **minimum spanning** tree problem is as follows:
  - **Given**: A weighted, undirected graph $G = (V, E, \omega)$.
  - **Compute**: a spanning tree of G with minimum cost (i.e., minimum total edge weight).
- For a given graph, the MST may not be unique.



Cost:
1+2+3+4+5+7+8=30

Prim's Algorithm, Informally



- Start by picking any node to add to "tree", T.
- While T is not a spanning tree, greedily add **lightest** edge from a node in T to a node not in T.
  - "Lightest" = edge with the smallest weight.
- **Is this guaranteed to work?** Yes, as we'll see.

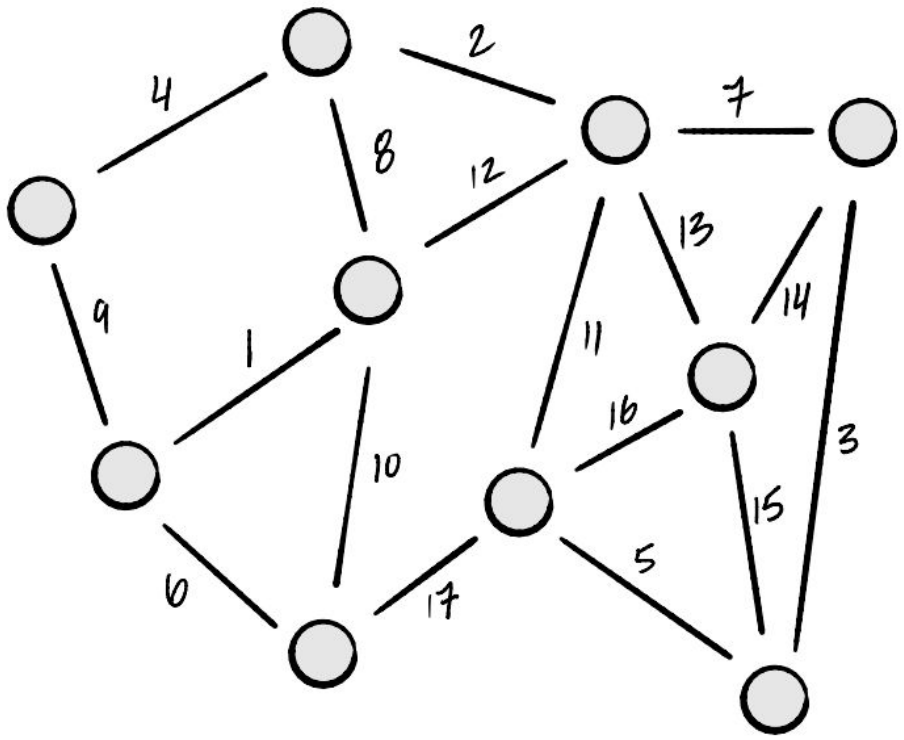
Last time: Building MSTs

- How do we build a MST efficiently?
- We'll adopt a **greedy** approach.
  - Build a tree edge-by-edge.
  - At every step, doing what looks best at the moment.
- This strategy isn't guaranteed to work in all of life's situations, but it works for building MSTs.

Last time: Two Greedy Approaches

- We'll look at two greedy algorithms:
  - **Last time**: Prim's Algorithm
  - ***Today: Kruskal's Algorithm***
- Differ in the order in which edges are added to tree.
- Also differ in time complexity.

Prim's Algorithm, Informally



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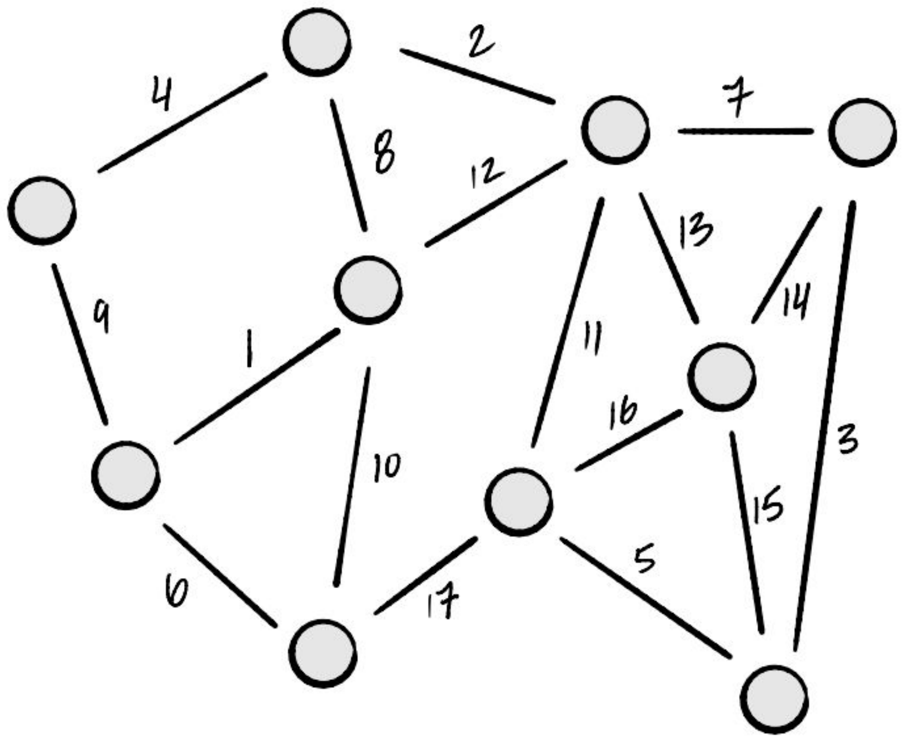
Kruskal's Algorithm, Informally



- Start with **empty** forest: $T = (V, E_{mst})$, where $E_{mst} = \emptyset$.
- Loop through **edges** in *increasing* order of weight.
  - If edge **does not** create a cycle in T, add it to T.
  - If T is a **spanning tree**, break.

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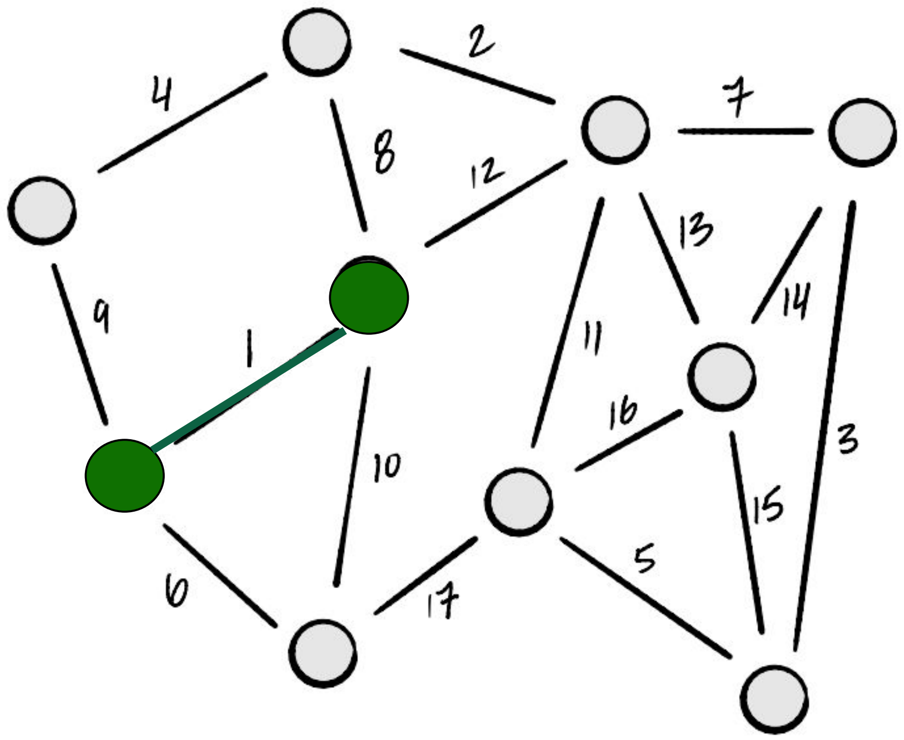

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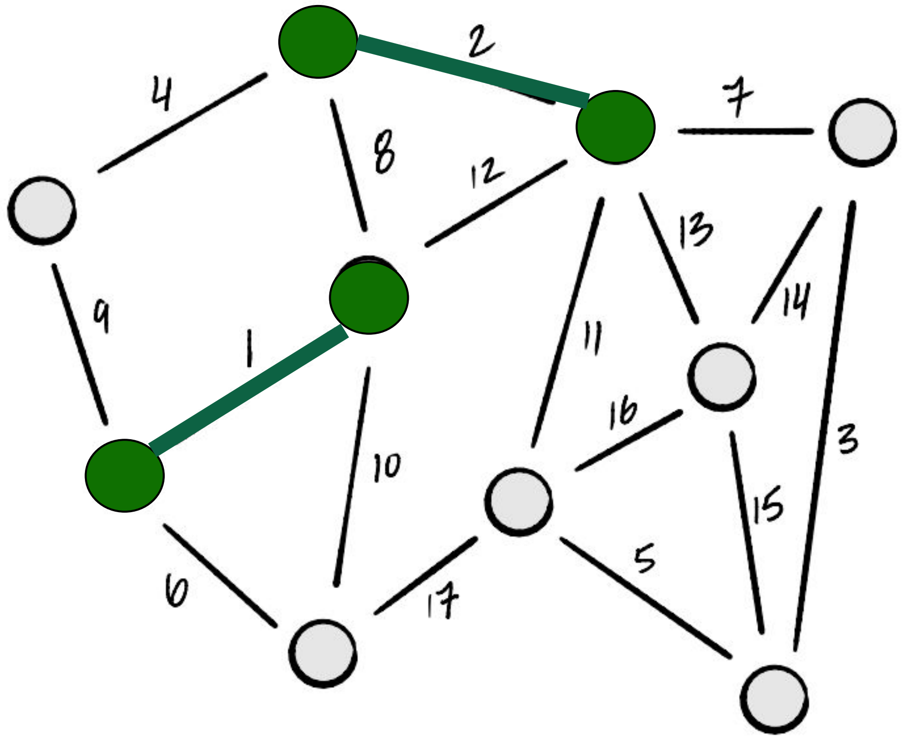

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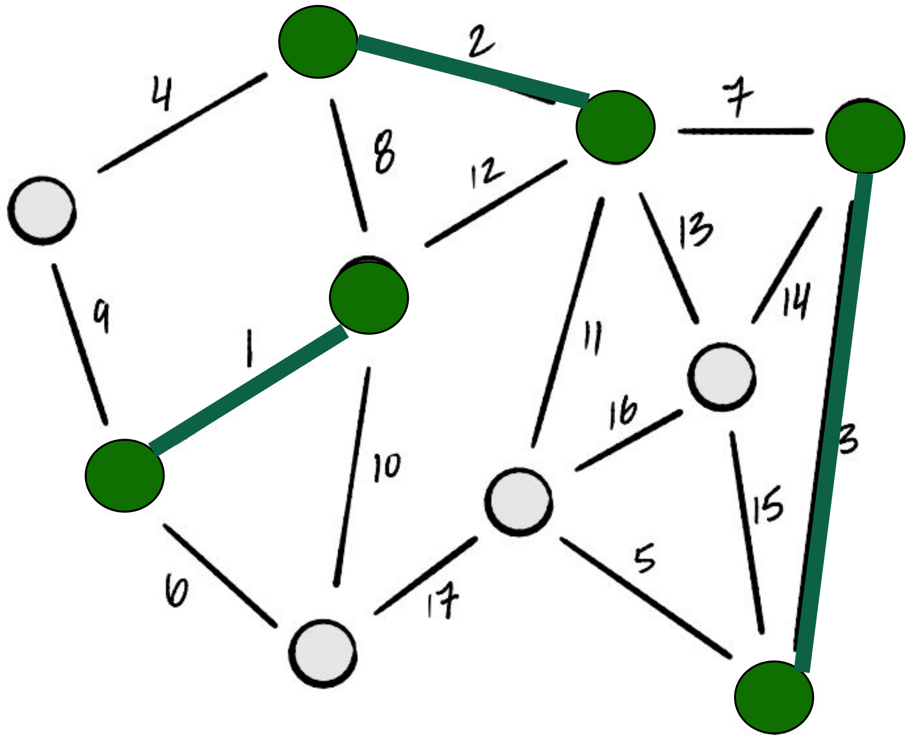

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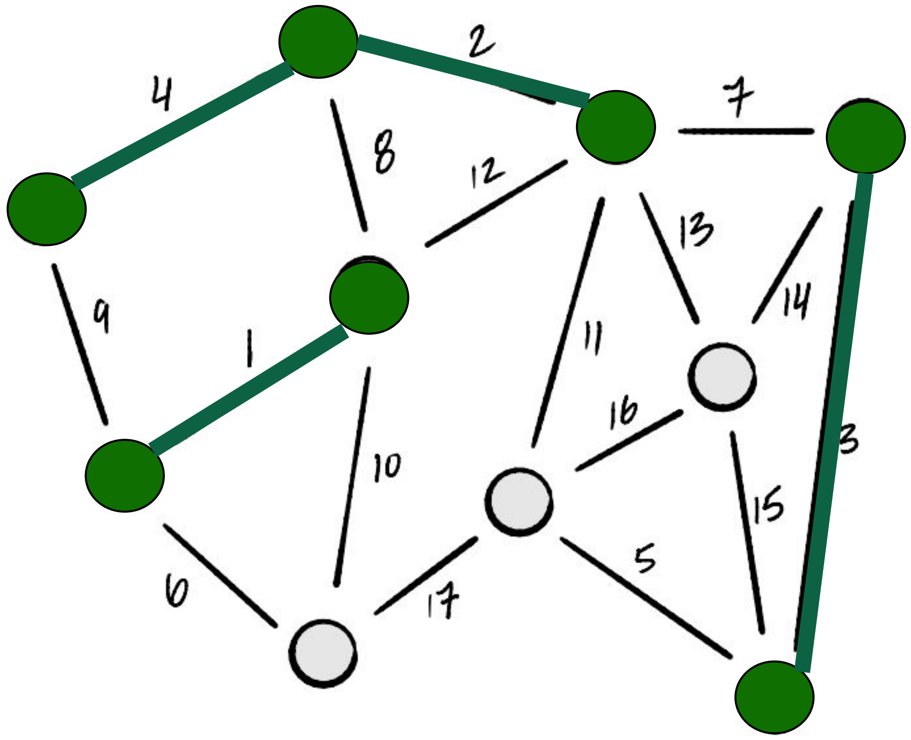

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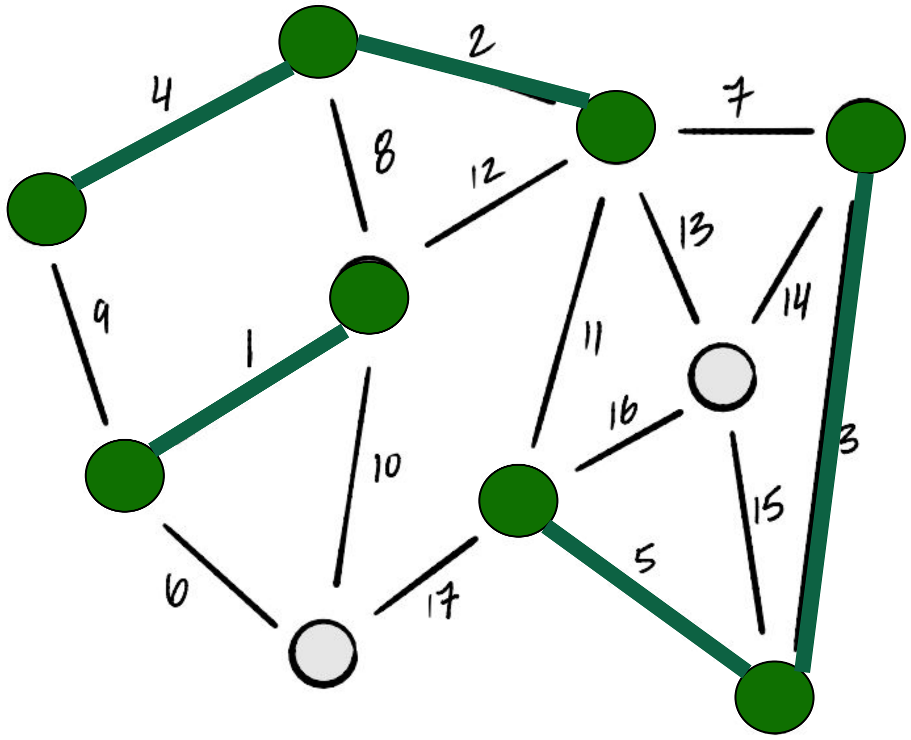

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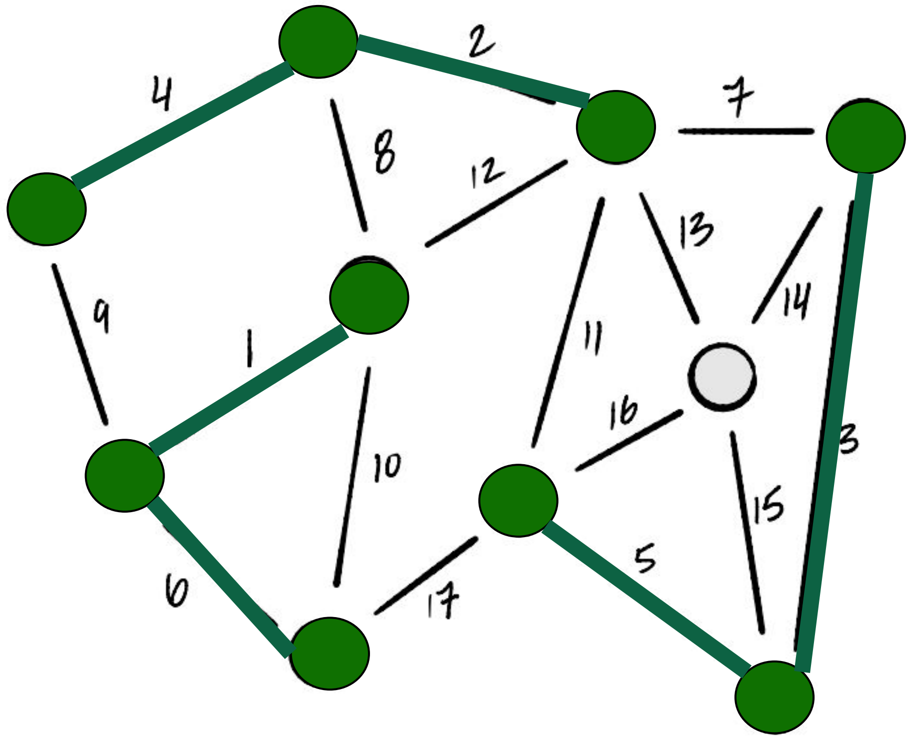

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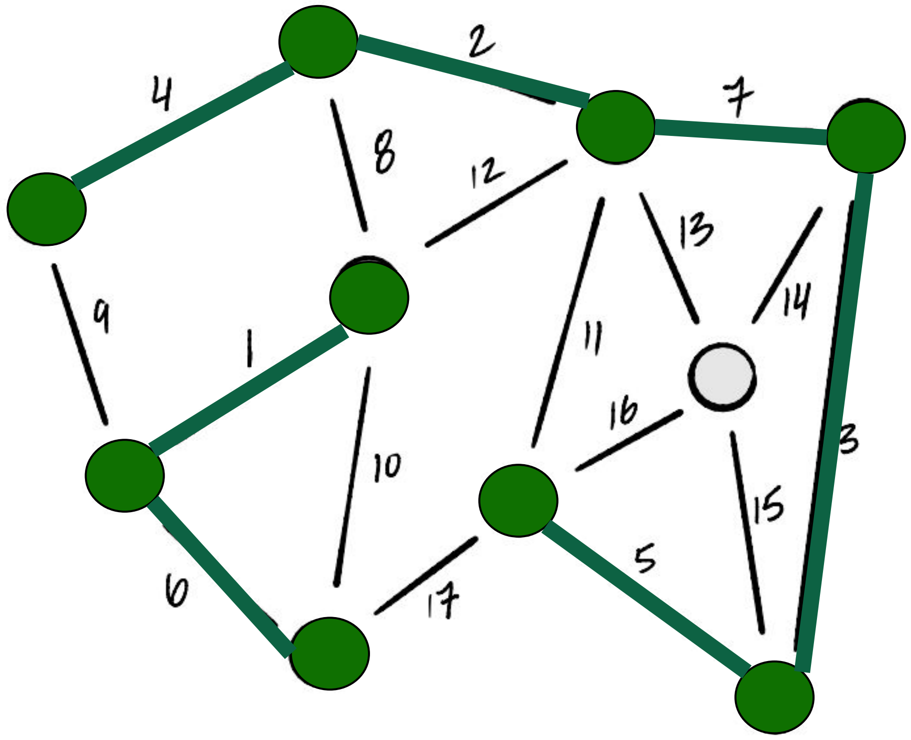

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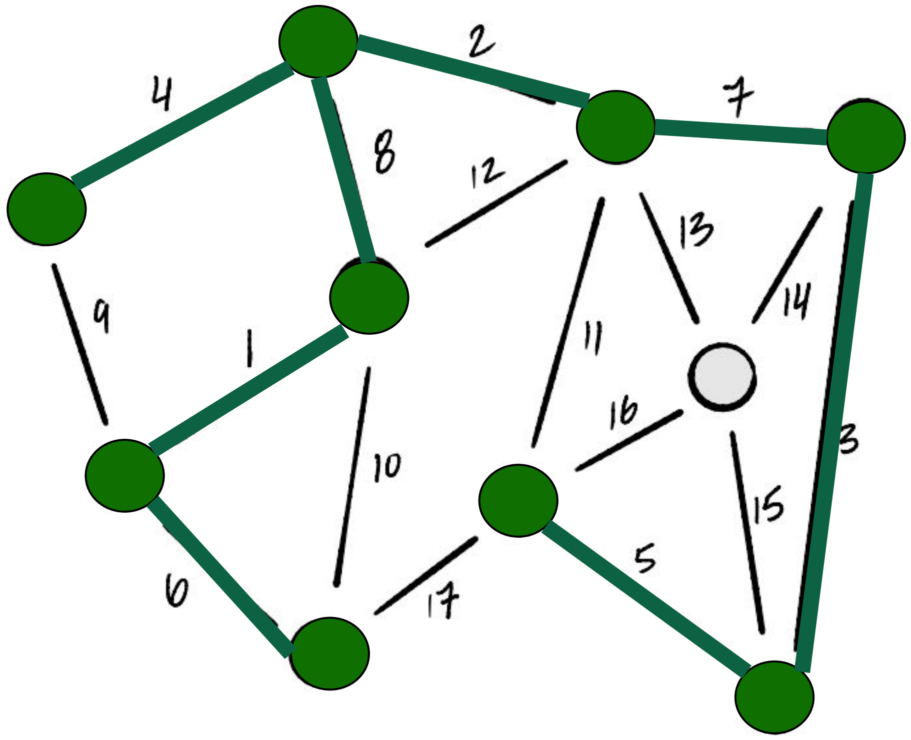

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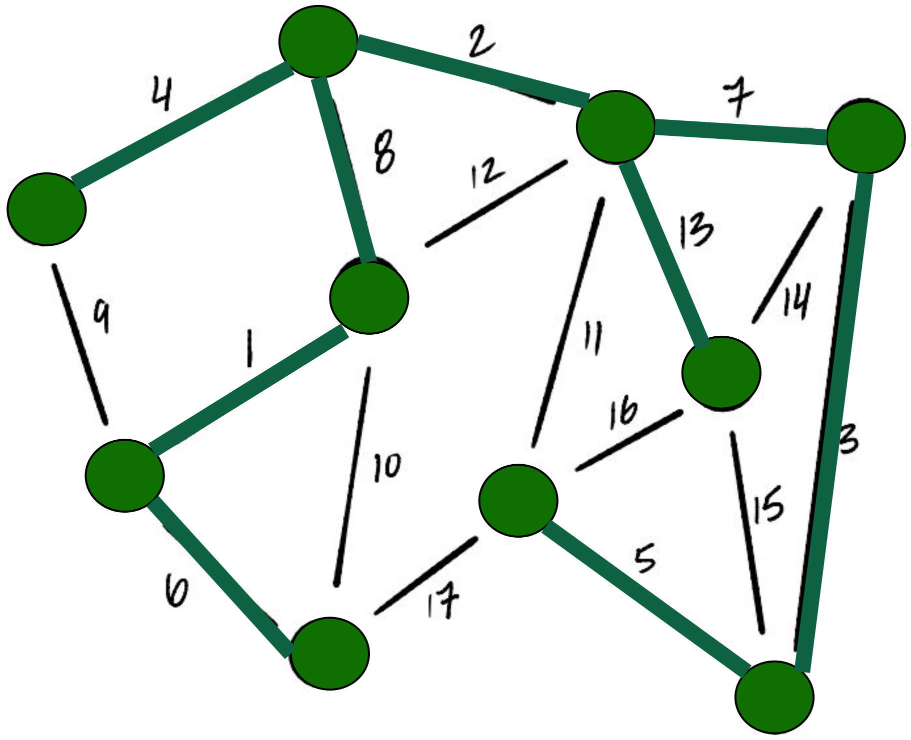

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Being Greedy

- **Prim**: add the **node** with smallest estimated cost and update neighbors.
  - Works locally, “grows” a connected tree.
- **Kruskal**: add the **edge** with smallest weight.
  - As long as it doesn’t make a cycle.
  - Edge can be anywhere in graph.

Kruskal's Algorithm (Pseudocode)

```
def kruskal(graph, weights):
    mst = UndirectedGraph()

    # sort edges in ascending order by weight
    sorted_edges = sorted(graph.edges, key=weights.get)

    for (u, v) in sorted_edges:
        # if u and v are not already connected
        if ...:
            mst.add_edge(u, v)

        # (optional) if mst is now a spanning tree, break
        if len(mst.edges) == len(graph.nodes) - 1:
            break

    return mst
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To check if u and v are connected

A: BFS

B: DFS

C: None

D: More than one

Checking for Connectivity

- Each iteration: check if u and v are connected in $T = (V, E_{mst})$.
- We *could* do a DFS/BFS on each iteration but . . .
  - $\Theta(V + E_{mst}) = \Theta(V)$ each time.
  - **Expensive!**
- **Remember:**
  - If you're computing something **once**, use a *fast algorithm*.
  - If you're computing it **repeatedly**, consider a *data structure*.

Disjoint Set Forests

- Represent a collection of disjoint sets.

{{1, 5, 6}, {2, 3}, {0}, {4}}

- `.union(x, y)`: Union the sets containing x and y.
- `.in_same_set(x, y)`: Return **True/False** if x and y are in the same set*.

**Usually implemented as a `.find(x)` method returning representative of set containing x.*

Example

```
>>> # create a DSF with {{0}, {1}, {2}, {3}, {4}, {5}}
>>> dsf = DisjointSetForest([0, 1, 2, 3, 4, 5])
>>> dsf.union(0, 3)
>>> dsf.union(1, 4)
>>> dsf.union(3, 1)
>>> dsf.union(2, 5)
>>> # dsf now represents {{0, 1, 3, 4}, {2, 5}}
>>> dsf.in_same_set(0, 3)
True
>>> dsf.in_same_set(0, 2)
False
```

Disjoint Set Forests

- Operations take $\Theta(\alpha(n))$ time, where n is number of objects in collection.
- $\alpha(n)$ is the **inverse** **Ackermann function.**
- It grows **very**, **very** slowly.
- *Essentially* constant time.

Disjoint Set Forests

- Can be used to keep track of Connected Components of a **dynamic graph.**
- Nodes of Connected Components are disjoint sets.
  - Add an edge (u, v): `.union(u, v)`
  - Check if u and v are connected: `.in_same_set(u, v)`
- To check if u, v are already connected:
  - BFS/DFS: $\mathbf{\Theta(V)}$ each time.
  - DSF: $\mathbf{\Theta(\alpha(V))}$ each time (essentially $\Theta(1)$).

```python
def kruskal(graph, weights):
    mst = UndirectedGraph()

    # place each node in its own disjoint set
    components = DisjointSetForest(graph.nodes)

    # sort edges in ascending order by weight
    sorted_edges = sorted(graph.edges, key=weights)

    for (u, v) in sorted_edges:
        if not components.in_same_set(u, v):
            mst.add_edge(u, v)
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Time Complexity

Θ(V)

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Time Complexity

Θ(V)

Θ(E log E)

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Time Complexity

$\Theta(V)$

$\Theta(E \log E)$

$O(E\ )$

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Time Complexity

$\Theta(V)$

$\Theta(E \log E)$

$O(E\ )$

$\alpha(V)$ per line

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Time Complexity

$\Theta(V)$

$\Theta(E \log E)$

$O(E\ )$

$\alpha(V)$ per line

$O(E\ \alpha(V)\ )$ for loop

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$\Theta(V)$

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$O(E)$

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$\Theta(V + E \log E + E\,\alpha(V))$

Time Complexity

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$\Theta(V)$

$\Theta(E \log E)$

$O(E)$

$\alpha(V)$ per line

$O(E\, \alpha(V))$ for loop

$\Theta(V + E \log E)$

Time Complexity

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$\Theta(V)$

$\Theta(E \log E)$

$O(E)$

$\alpha(V)$ per line

$O(E\ \alpha(V))$ for loop

$\Theta(E \log E)$ if assume connected

Time Complexity

- Assume graph is connected. Then $E = \Omega(V)$.
  - E >= V - 1
- Kruskal's takes $\Theta(E \log E) = \Theta(E \log V)$ time.
  - Dominated by sorting the edges.
- **Note**: if graph disconnected, Kruskal's produces a **minimum spanning forest**.

Kruskal vs. Prim

Kruskal v. Prim

- Both algorithms for computing MSTs.
- Which is “better”?
- There’s no clear winner.

Time Complexity

- **Prim**:
  - Binary heap: $\Theta(V \log V + E \log V)$
  - Fibonacci heap: $\Theta(V \log V + E)$
- **Kruskal**: $\Theta(E \log V)$
- If the graph is dense, $E = \Theta(V^2)$, and Prim's with Fibonacci heap "wins".
  - $\Theta(V^2)$ versus $\Theta(V^2 \log V)$.

Not so fast...

- Fibonacci heaps are hard to implement, **high** overhead.
- Prim's will be faster for **very large dense graphs**.
- But Kruskal's may be faster for **smaller dense graphs**.
- The right choice depends on your application.

Main Idea

- Asymptotic time complexity isn't ***everything***.
- For small inputs, the "inefficient" algorithm ***may*** beat the "efficient" one.
- There's also ease of implementation to consider.

Quick check

You run Kruskal's algorithm on a connected, weighted graph with V vertices and E edges.

During execution, you notice that exactly k edges are skipped because they would have formed cycles. Which of the following must be true?

A. The graph contains exactly k cycles.

B. The number of connected components in the graph at the start of the algorithm was k + 1.

C. The graph has at least k extra edges beyond what is needed for a spanning tree.

D. The number of edges with weight ties (equal weights) is at least k.

Did everything make sense about Kruskals?

A: Yes: 100%

B: 80%

C: 60%

D: 40%

E: No :(

MSTs and Clustering

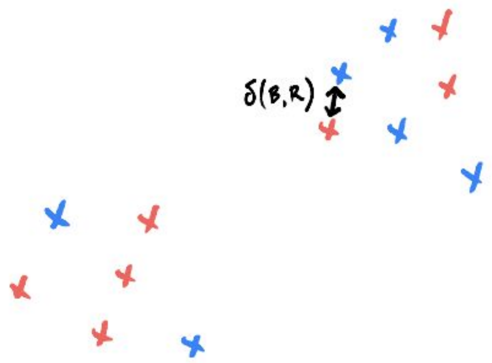
Clustering: Find Groups

Goal: identify the groups in data.

Example:

Clustering, Formalized

- We frame as an **optimization** problem.
- **Given**: n data points.
- **Goal**: assign color to each point (red or blue) to *maximize* the distance between the *closest* pair of red and blue points.



Bad Clustering

Clustering, Formalized

- We frame as an **optimization** problem.
- **Given**: n data points.
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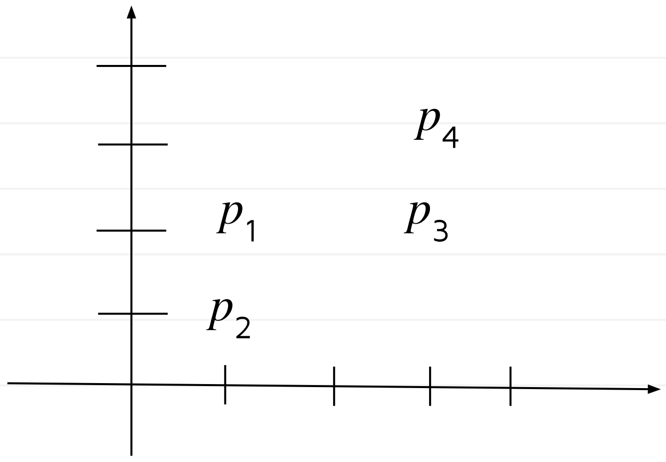
Brute Force Solution

- Try all possible assignments; return best.
- If there are n data points, there are $\Theta(2^n)$ assignments.
- **Exponential time**; very slow.
    - Practical only for ~ 50 data points.
- Instead, we will turn it into a **graph problem**.

Distance Graphs

- Given n data points, $p_1, p_2, \dots, p_n$, create *complete* graph with $V = \{p_1, \dots, p_n\}$.
- Set weight of edge (p_i, p_j) = $\mathrm{dist}(p_i, p_j)$.
- The result is a weighted, undirected **distance graph.**



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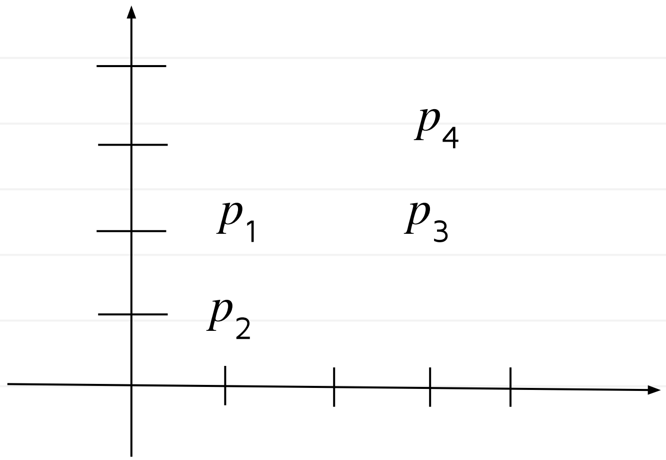

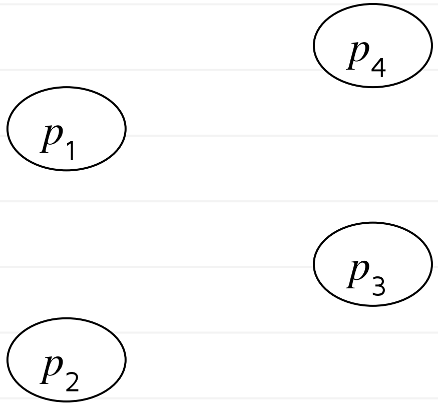

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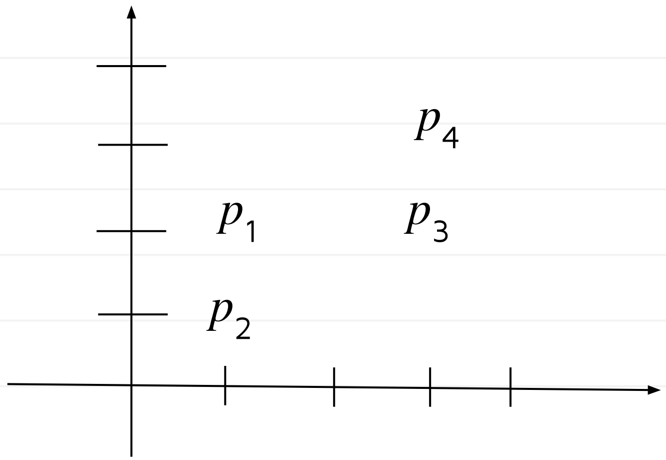

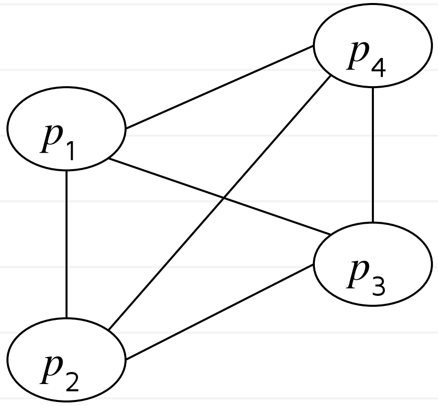

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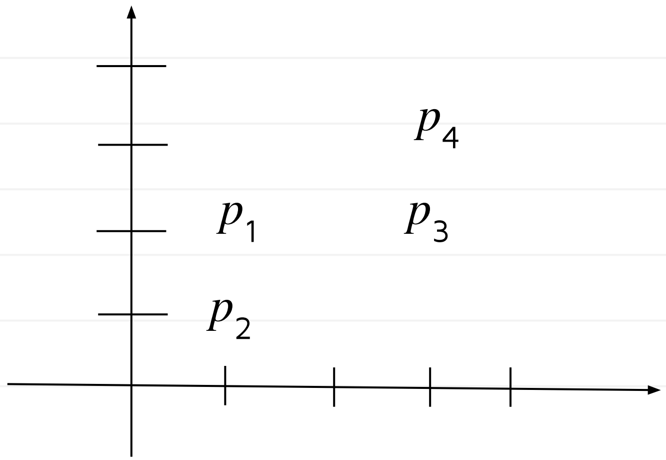

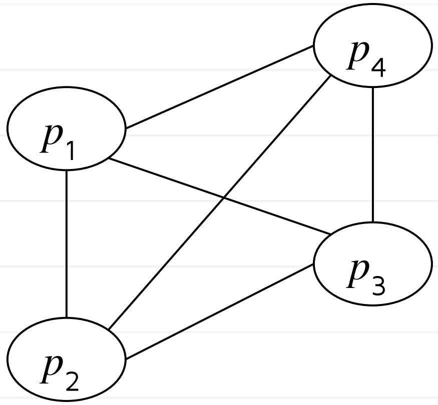

Weight is the distance

Distance Graphs

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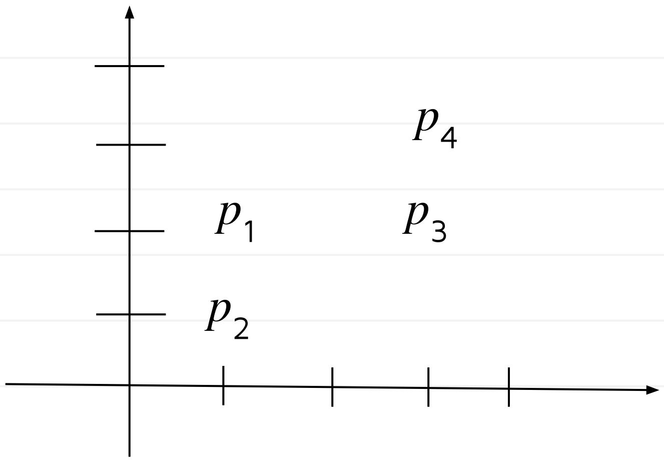

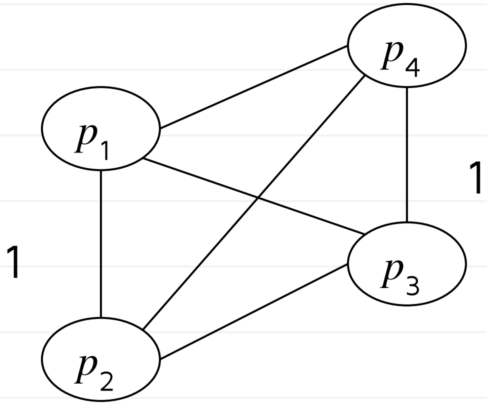

Weight is the distance

Distance Graphs

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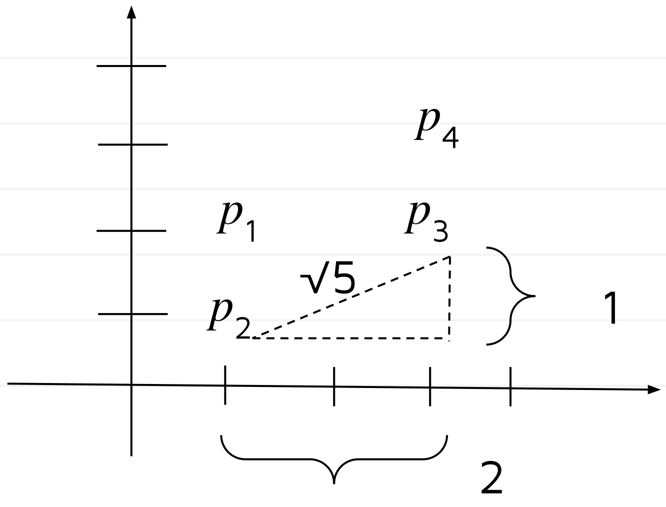

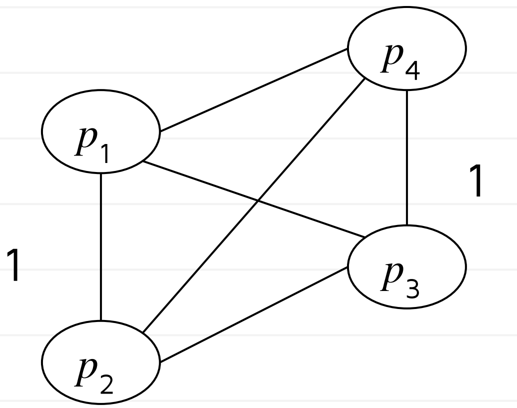

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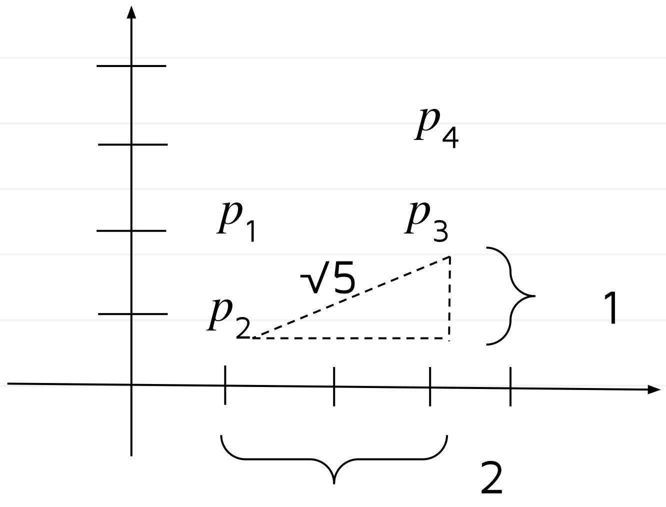

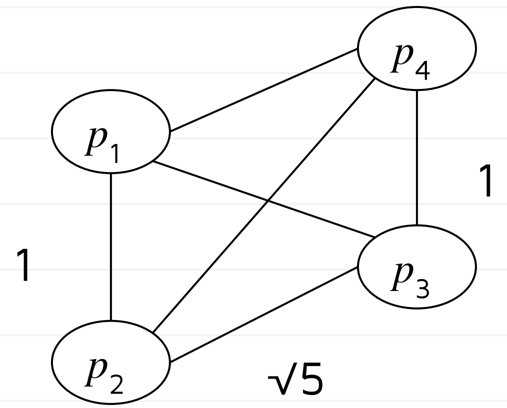

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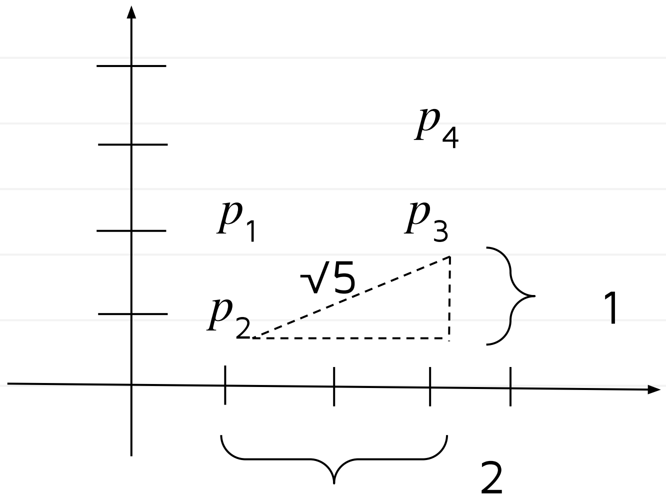

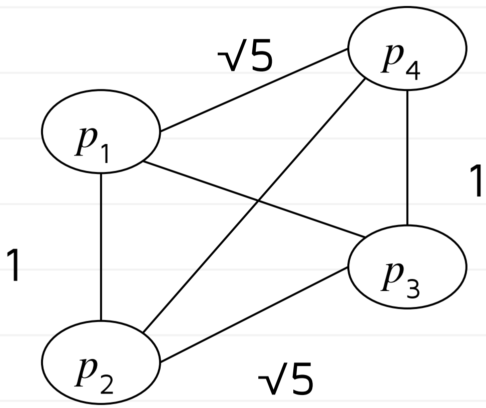

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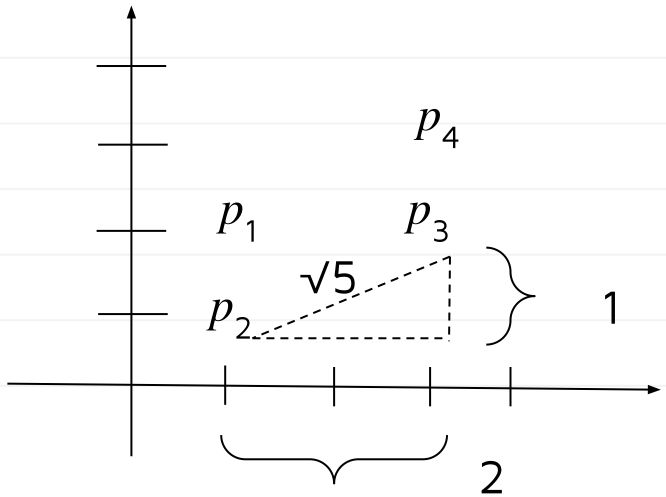
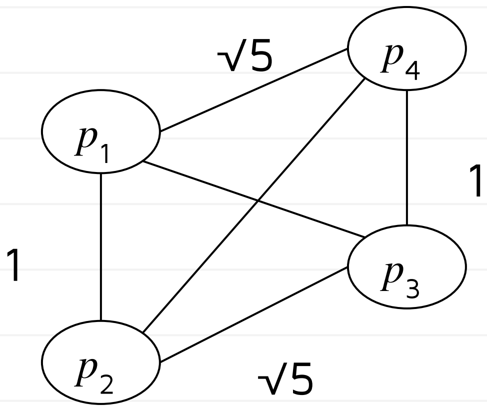




You can finish the rest

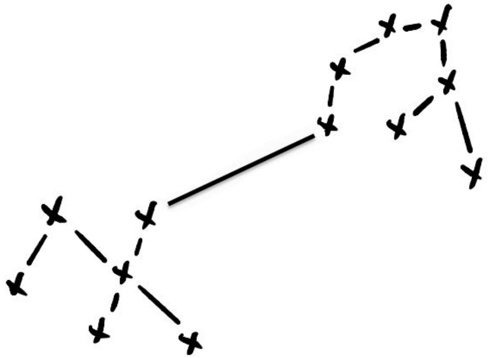
Main Idea

We can always think of a set of points in a (metric) space as a weighted distance graph.

This is a ***very important idea,*** because it allows us to use our graph algorithms!

Clustering with MSTs

- Given n data points and a number of clusters, k:
  - Create distance graph G.
  - Run Kruskal's Algorithm on G until there are only k components.



- The resulting connected components are the clusters.
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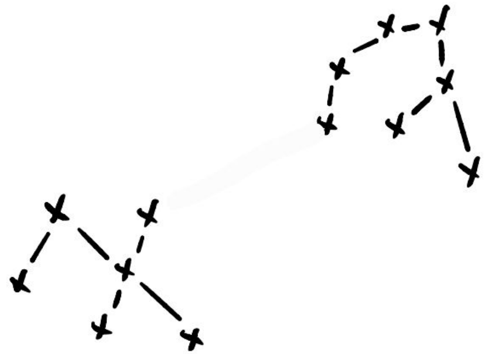

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Single-Linkage Clustering

- Time complexity of single-linkage is determined by Kruskal's Algorithm: $\Theta(E \log E)$.
- Since distance graph is complete, $E = \Theta(V^2)$, and so $\Theta(E \log E) = \Theta(V^2 \log V) = \Theta(n^2 \log n)$
- Practically, can cluster ~ 10, 000 points.

Summary

- We started the quarter with a **brute force solution**.
  - Took $\Theta(2^n)$ time, only feasible for a few dozen points.
- We've now reframed the problem using graph theory.
  - Now only $\Theta(n^2 \log n)$ time!
  - Feasible for tens of thousands of points.

Why Algorithms?

- Data scientists use computers as tools.
- But solving a problem isn’t just about coding it up.
- You need to know how to analyze your code and use the right algorithms and data structures to make your solution efficient.

Thank you!

Do you have any questions?

CampusWire!